

KAKE, AK, USA

WMO: 703855

Lat: 56.961N

Lon: 133.910W

Elev: 172

StdP: 14.60

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25369

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	17.4	0.0	5.5	17.7	2.9	6.4	20.3	22.1	38.7	19.7	39.6	6.1	70	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	69.7	57.8	65.7	56.8	63.3	55.1	59.4	66.3	58.0	63.8	56.7	61.7	5.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.8	66.8	61.4	54.9	64.4	60.3	54.1	62.8	59.6	25.9	66.8	25.1	63.7	24.2	61.6	66.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	14.8	7.4	75.3	4.1	4.6	4.5	78.6	2.1	81.3	-0.2	83.9	-3.2	87.2		
(5)			6.4	61.4	3.8	1.7	3.7	62.7	1.5	63.7	-0.7	64.7	-3.4	66.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	32.9	33.2	35.2	40.7	47.4	52.6	55.9	55.7	50.9	43.5	36.9	33.1
	DBStd	10.08	7.44	6.44	5.71	3.86	4.52	3.76	3.06	3.19	3.52	4.59	6.17	6.94
	HDD50	3013	531	471	460	280	107	11	0	1	29	205	393	525
	HDD65	7957	996	891	925	729	547	374	284	288	423	667	843	990
	CDD50	533	0	0	0	1	25	88	182	178	56	3	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	14	0	0	0	0	0	6	6	2	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	6.6	7.3	7.0	7.4	6.8	6.0	5.0	4.9	5.0	6.2	7.2	7.9	8.0
Precipitation	PrecAvg	104.8	11.5	8.6	8.4	6.4	4.6	3.8	4.8	6.7	10.7	13.4	12.9	13.0
	PrecMax	142.7	21.4	17.8	15.8	9.8	11.5	7.0	8.9	14.6	19.3	20.9	23.1	28.2
	PrecMin	75.3	3.5	3.1	3.0	0.6	0.3	1.4	0.4	1.7	3.9	5.9	5.9	5.9
	PrecStd	17.0	3.6	3.9	3.3	2.3	2.6	1.4	2.0	3.0	3.7	3.4	4.1	5.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	48.0	51.7	59.7	69.7	73.4	72.9	72.7	65.3	55.4	51.9	48.1
		MCWB	42.9	42.0	44.2	46.8	53.1	59.1	60.5	58.8	54.4	51.0	46.0	44.5
	2%	DB	45.5	45.3	46.5	54.2	63.1	67.6	68.2	67.6	61.1	54.1	48.2	46.0
		MCWB	42.4	41.3	42.0	45.2	51.8	56.7	58.9	58.4	53.1	50.3	45.9	43.4
	5%	DB	43.4	43.4	45.1	50.5	59.2	63.4	64.4	64.3	58.3	52.1	46.2	44.8
		MCWB	40.1	39.7	40.7	43.5	49.5	53.9	56.7	56.7	53.2	48.7	43.7	42.1
	10%	DB	42.4	42.1	43.2	47.9	55.3	60.7	62.8	62.7	56.6	50.1	45.1	42.7
		MCWB	39.6	39.1	38.9	42.4	48.3	52.7	55.7	56.0	52.5	47.1	42.2	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	44.3	46.0	48.6	54.9	60.2	61.5	61.3	57.8	53.1	48.2	45.9
		MCDB	46.7	46.0	49.6	56.4	63.9	71.6	70.1	69.7	60.7	54.8	49.6	46.6
	2%	WB	42.9	41.9	43.5	46.3	52.7	57.2	59.3	59.5	55.8	50.7	46.2	44.3
		MCDB	45.1	44.9	46.2	52.1	60.5	65.3	66.3	65.3	59.0	53.1	47.6	45.8
	5%	WB	41.0	40.5	41.6	44.5	51.0	55.1	57.9	58.2	54.3	48.9	44.5	41.9
		MCDB	43.9	43.7	44.9	49.1	57.1	61.5	63.7	63.3	57.1	51.2	46.2	44.5
	10%	WB	39.5	39.2	39.8	43.0	49.6	53.4	56.5	56.8	53.1	47.1	42.3	40.0
		MCDB	42.4	42.1	43.1	47.1	55.2	59.0	61.5	61.4	55.7	49.2	44.9	42.9
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.6	9.9	12.8	13.7	12.4	11.2	11.5	10.4	8.6	7.1	6.9
		MCDBR	7.8	9.3	12.7	18.2	21.9	20.0	17.7	17.4	13.8	10.3	7.9	7.6
		MCWBR	6.5	7.2	9.1	11.3	12.0	10.9	9.8	10.2	8.7	7.9	6.4	6.3
	5% WB	MCDBR	7.7	8.3	11.7	15.3	17.2	17.0	15.3	14.4	10.3	8.3	7.1	7.0
		MCWBR	6.7	6.7	8.8	10.2	10.2	9.5	9.0	8.9	7.4	7.0	6.2	6.1
Clear-Sky Solar Irradiance	taub		0.293	0.317	0.325	0.377	0.371	0.374	0.379	0.367	0.357	0.330	0.303	0.277
	taud		2.271	2.346	2.405	2.271	2.391	2.427	2.450	2.506	2.489	2.461	2.285	2.208
	Ebn at Noon		186	234	266	265	275	275	271	267	252	228	179	155
	Edh at Noon		16	23	27	36	34	34	32	29	25	20	16	13
All-Sky Solar Radiation	RadAvg		140	382	757	1239	1556	1557	1341	1129	737	403	181	100
	RadStd		16	63	92	107	151	137	152	133	110	52	24	14

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page